Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/17/2023
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	03/01/2016	**End Date:**	02/28/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 2/28/2023 per NSSC (Ed., 1/21/22) NOTE: End date changed to 2/28/2022 per PI (Ed., 1/22/21)		
Key Personnel Changes/Previous PI:	Dr. Chengyao Li, research assistant professor at Case Western Reserve University became a Co-investigator of this project in February 2022. He was previously a post-doc associated with this project.		
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center) Johnston, Michael C Ph.D. (Universities Space Research Association) Li, Chengyao (Case Western Reserve University)		
Grant/Contract No.:	NNX16AE84A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A. [Note: Grant NNX10AD99A has been renamed, "Research Opportunities in Combustion Science," per NSSC information (Ed., 1/5/24).] Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, and shoulder and wake regions results in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with five embedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The microgravity experiment using the Combustion Integrated Rack (CIR) in the International Space Station (ISS) has started in February 2023. It is expected to continue until June 2023 followed by data analysis under a renewed grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	In this past year, our major effort is to prepare the Growth and Extinction Limit (GEL) microgravity experiment in the International Space Station (ISS). The progress in this period includes ignition and flame-quenching extinction. Though these works were restricted either by normal gravity buoyancy-induced flow or limited experiment duration (5.2 s) in the drop tower, supplemented by numerical model and theoretical analysis, they still provided significant data to prepare for the upcoming ISS GEL experiment. Ignition delay and ignition boundary in normal gravity In this work, the ignition limits and the ignition delay times of the GEL sphere were investigated with the 10 mm – B type SoFIE (Solid Fuel Ignition and Extinction) igniter in normal gravity. The igniter wire's electrical current and power-on time, its distance from the sample surface, the forced flow velocity, oxygen percentage, and pressure were systematically varied. Two ignition delay times were determined: igniter-assisted ignition and self-sustained ignition. The solid temperature profiles were measured and used to determine the sub-surface temperature gradient at ignition. A numerical model was developed to study the heat transfer mechanism between the igniter and the solid surface during the pre-pyrolysis heat-up period. In addition to its scientific value in fundamental research of blunt body Polymethylmethacrylate (PMMA) samples' ignition with heated wire, by using the equivalent flow strain rate, this research also provided valuable data in the preparation of the GEL experiment. For the ISS experiments, it is important to ensure that the sample can be ignited, but on the other hand, the igniter is not to overheat the sample leading to a too strong initial flame that shortens the flame growth period. To understand the details and the mechanisms of heat transfer between the igniter and the solid sample in the preheating phase, a numerical model was developed and solved with ANSYS Fluent to study the fluid-thermal processes between the igniter and the solid sample. The heat transfer modes include convection and radiation between the igniter and the sample (solid-to-solid) and heat conduction inside the solid. Pyrolysis is ignored during this pre-ignition period. The computation shows that at the stagnation point, radiative heat transfer is, in general, more important than convective heat transfer, and constitutes about 60% of the net total heat flux. The numerical model was also used to predict the ignition delay and heat transfer under microgravity. The numerical model was also used to predict the ignition delay and heat transfer under microgravity. In the absence of gravity-induced buoyant flow, the convective heat transfer is weaker but the heat flux at the stagnation point region gets stronger. It leads to shorter time delay to heat the solid surface to the pyrolysis temperature. Another utilization of the normal gravity data for the ISS GEL microgravity experiment is to employ the equivalent stretch rate concept. For example, the 5 cm/s flow rate at normal gravity is equivalent to a 26 cm/s pure forced flow rate at microgravity. This provides guidance in the microgravity experiments. A complete document of the normal gravity ignition work can be found in: Li C, T'ien JS, Johnston MC, Olson SL, Ferkul PV. "Ignition of thermally thick blunt body PMMA samples using a heated wire." Fire Saf. J. 2022 Oct;133:103663. https:// Dynamic response and induced quenching extinction in normal to zero gravity transition The transient response of the diffusion flame over the stagnation region of the GEL sphere to an abrupt transition to zero-gravity was investigated experimentally in the NASA Glenn Research Center (GRC) 5s drop tower. A corresponding numerical modeling effort was carried out by Professor Sheng-Yen Hsu and his student Ching-Wei Wu. One of the objectives of this effort is to locate the flame extinction boundary during the 1g-0g transition. These data, although transient, can help us to locate the range of parameters in the determination of the quenching limits in the ISS GEL experiment.

	In this experiment, an initial flame was established in normal gravity in a 17% oxygen atmosphere before the sudden release in a 5.18-second drop tower for zero gravity. Two parameters were investigated: the forced flow velocity and the degree of preheating in the surface layer of the GEL sample. The amount of preheating was controlled by monitoring the embedded thermocouples reading and adjusting the time of normal gravity burning before releasing to microgravity. The experimental results showed that the flame tip retreats by the end of the 5.18-second drop, and the flame may either survive or die depending on the controlling parameters. The experiments also showed rapid and dynamic flame changes, both in shape and color, in response to the changes of flow patterns when the buoyant driving force is suddenly removed. A numerical model was developed and implemented with ANSYS Fluent to gain more insight on the transient flow pattern and the solid-gas energy balance at the solid surface during the transition. The model solves full two-dimensional transient Navier-Stokes momentum, species, and energy equations with direct numerical simulation. Zeroth-order Arrhenius reaction is assumed for the solid pyrolysis at the solid surface. The produced Methyl Methacrylate (MMA) monomer reacts with oxygen in the gas phase through a one-step overall combustion reaction. The solid phase solves the two-dimensional transient heat conduction. Gas-phase radiation is ignored but solid surface radiation is considered. Solid phase is coupled with the gas phase though solid surface energy balance. Ignition is through external energy input in a small volume (0.0264 cm3) in the upstream forward stagnation point region with a total heat input of 2.64 W in 10 s. An extinction map is presented using the experimental and the numerical data. Most of the work has been carried out previously but the results have been written up in this period. A manuscript has been submitted for journal publication and is under review. Preparation of test samples for ISS experiment Custom made 4-cm diameter cast PMMA sphere samples with 5 embedded thermocouples have been provided to NASA. All samples have been CT x-rayed in the Case Western Reserve University (CWRU) Case Center for Imaging Research to determine the positions of each thermocouple. The measured solid in-depth temperatures are to be used to evaluate the sub-surface temperature gradient.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Li C, T'ien JS, Johnston MC, Olson SL, Ferkul PV. "Ignition of thermally thick blunt body PMMA samples using a heated wire." Fire Saf. J. 2022 Oct;133:103663. https://doi.org/10.1016/j.firesaf.2022.103663 , Oct-2022
Articles in Peer-reviewed Journals	Dietrich DL, T'ien JS. "Can cool flames support quasi-steady droplet burning? " Combust. Flame. 2022 Nov;245:112367. https://doi.org/10.1016/j.combustflame.2012.07.012 , Nov-2022
Articles in Peer-reviewed Journals	Johnston MC, T'ien JS, Hsu S-Y, Wu C-W, Olson SL, Ferkul PV. "Quenching extinction of solid sphere diffusion flames induced by a sudden removal of gravity." Fire Safety Journal. 2024 Jun 1;146:104137. https://doi.org/10.1016/j.firesaf.2024.104137 , Jun-2024